

Client Change Request (CCR)**CCR Number:** CCR-006**Client:** Sweet Delights Bakery**Submitted By:** Emily Thompson, Bakery Owner**Priority:** Medium**Requested Delivery:** End of Today's Sprint**Assigned Developer:** _____**Date Assigned:** _____

Business Background

Sweet Delights Bakery receives dozens of customer orders every morning. Employees currently calculate taxes, discounts, and final prices using handheld calculators. As the number of orders has increased, the bakery has found that these calculations are becoming more time-consuming and prone to errors.

The bakery owner would like a computer program that calculates the total amount due for each order. Because the pricing rules may change over time, the owner wants each calculation to be performed independently so that individual pricing rules can be updated without affecting the rest of the program.

At this time, the bakery only needs to calculate one customer order each time the program is run.

Customer Requirements

The employee should be able to enter the total cost of a customer's order before taxes and discounts.

The program should perform separate calculations for:

- The sales tax
- Any customer discount
- The final amount due

The main program should use these individual calculations to produce the final receipt.

The receipt should clearly display:

- Original order amount
- Discount amount
- Sales tax
- Final amount due

The receipt should be neatly formatted and easy for both employees and customers to understand.

Acceptance Criteria

The customer will consider the software complete when all of the following are true.

- ☐ Employee can enter the original order amount.
 - ☐ Program calculates the discount.
 - ☐ Program calculates the sales tax.
 - ☐ Program calculates the final amount due.
 - ☐ Program displays all calculated values.
 - ☐ Program produces a neat and readable receipt.
 - ☐ Program successfully calculates multiple test cases provided by the customer.
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Out of Scope

The customer is **not** requesting:

- Processing multiple customer orders
- Saving orders to a file
- Printing receipts
- Accepting customer names
- Inventory management
- Multiple tax rates
- Loyalty programs
- Coupons
- Credit card processing
- Sales reports

If these features appear in the software, they will not be evaluated.

Example Scenario 1

Employee Input

Order Amount: \$25.00

Expected Output

Sweet Delights Bakery

Customer Receipt

Original Order : \$25.00

Discount : \$2.50

Sales Tax : \$1.58

Final Total : \$24.08

Example Scenario 2

Employee Input

Order Amount: \$12.50

Expected Output

Sweet Delights Bakery

Customer Receipt

Original Order : \$12.50

Discount : \$1.25

Sales Tax : \$0.79

Final Total : \$12.04

Notes from the Customer

"Our current calculator works, but every time we change our tax rate or decide to offer a new discount, we have to rethink the entire calculation process. I'd like the program to be organized so that each calculation is handled independently. That way, if one rule changes in the future, only that part of the program needs to be updated."

Developer Notes

This section intentionally left blank.

Students will complete the software requirements during class before beginning implementation.

Questions for the Customer

Before writing any code, list any questions you would ask the customer to clarify the request.

Examples:

- 1.

Reminder

The customer has intentionally described the problem using **business language**, not programming terminology. During class, your first task is to translate these customer requirements into **software requirements** before writing any C code.

For this assignment, you should use only the C language features that have been covered through **Lesson 5**, including:

- Variables and constants
- Input/output
- Selection statements (if and switch)
- Loops
- Functions
- Basic arithmetic
- Formatted output

Do **not** use C language features that have not yet been introduced in the course.